



BOILEAU GUILLAUME

Ingénieur Moyens d'Essais | Bancs de test, acquisition, LabVIEW, signal & validation

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EXPERIENCE

Software Engineer - System Integration, Hardware Interfaces & Test Diagnostics

VEV Platform Services France

📅 2025 - 2026

📍 Valbonne, France

Context: Development and integration of software services for an EV charging CPMS platform connected to operational infrastructure, charging stations, data flows and hardware systems.

Objective: Support reliable software integration, operational data-flow continuity, hardware/software interface diagnostics and test activities on charging infrastructure.

- **Software development:** Developed and maintained web and backend services using TypeScript, Angular and Node.js in an operational industrial environment.
- **System integration:** Contributed to the integration and maintenance of software services interacting with operational infrastructure and connected equipment.
- **Hardware/software interfaces:** Analysed interactions between platform services, connected charging stations, data exchanges, electrical circuits, communication buses and operational behaviours.
- **Electrical test support:** Worked on charging-station interfaces, circuits and buses used for testing, diagnostics and functional checks.
- **Field interventions:** Performed hands-on maintenance on EV charging stations for test activities, including soldering, electrical checks and support to integration tests.
- **Electrical safety:** Held BO / BE Essais electrical authorization for test-related activities on charging infrastructure.
- **Data flows:** Implemented, monitored and corrected FTP-based operational data exchange services.
- **Issue analysis:** Investigated incidents involving software behaviour, operational data exchanges, interface continuity and hardware-connected systems.
- **Coordination:** Worked with technical stakeholders to clarify issues, follow priorities and support iterative delivery.
- **Tooling:** Used Zenhub, Git-based workflows, Confluence, OpenSearch and Sentry for delivery follow-up, diagnostics and technical documentation.

Senior R&D Engineer - Test Logic, Signal Processing, Validation & Technical Coordination

CNES, Laboratoire Lagrange, Observatoire de la Côte d'Azur

📅 2023 - 2025

📍 Nice, France

Context: Engineering and data-analysis activities for the LISA ESA/NASA space mission, involving complex software chains, signal analysis, simulation workflows, validation campaigns and international coordination.

Objective: Develop, integrate and validate Python-based analysis workflows under strong consistency, traceability, performance and verification constraints.

- **Validation workflows:** Structured comparison, verification and validation logic for heterogeneous simulation outputs and technical deliverables.
- **Software platform:** Developed CATpyLISA, a Python platform for large-scale model integration, comparison, validation and technical review. [GitLab: CATpyLISA](#)
- **Signal processing:** Analysed weak, noisy and superposed signals using spectral methods, signal/noise separation logic and performance metrics.
- **Measurement-oriented analysis:** Worked on simulated and instrument-related data where performance depends on noise, sensitivity, acquisition quality and signal interpretation.
- **Automated test logic:** Defined repeatable analysis sequences, comparison rules, consistency checks and acceptance-oriented diagnostics.
- **Critical systems context:** Worked in a high-rigor space mission environment involving traceability, technical evidence, review logic and mission-level performance objectives.
- **Technical studies:** Analysed sensitivity to modelling assumptions, numerical parameters, data-processing choices and configuration changes.
- **Data reliability:** Ensured traceability of inputs, configurations, outputs, assumptions, metadata and validation results.
- **Technical reporting:** Produced analysis notes, validation summaries, technical documentation and review material for contributors and project stakeholders.
- **Coordination:** Coordinated contributors, supervised interns and supported technical activities across multiple stakeholders in an international environment.

Data Scientist - Noise Diagnostics, Measurement Analysis & Performance Validation

Universiteit Antwerpen, Belgium

📅 2021 - 2023

📍 Antwerp, Belgium

Context: Data analysis and performance studies for precision instruments in a high-requirement international environment linked to gravitational-wave detector performance.

Objective: Identify, characterize and mitigate factors affecting measurement quality and system sensitivity using statistical modelling, signal processing and validation-oriented methods.

PROFILE FOR TERX

Engineer at the interface between test means, software development, signal processing, acquisition-oriented data analysis and validation of complex systems. Background developed through space mission engineering, precision instrumentation and industrial software environments, complemented by hands-on exposure to electrical test activities on EV charging infrastructure.

Experienced in **Python**, **LabVIEW**, signal processing, data acquisition logic, measurement analysis, automated validation workflows, hardware/software interfaces, electrical test support, technical documentation and operational diagnostics. Familiar with **LabVIEW** from academic training and able to reactivate and deepen it in an industrial test-bench context.

For an **Ingénieur Développement Moyens d'Essais** role in an industrial environment, I bring a strong engineering profile: understanding physical measurements, structuring automated test logic, analysing signal and acquisition quality, validating system performance, documenting results and working across software, hardware, production and engineering constraints.

FIT FOR TEST MEANS / INSTRUMENTATION

- Strong background in signal processing, measurement analysis, noise characterization and performance assessment.
- Hands-on experience with Python-based analysis workflows, automated validation logic, consistency checks and technical reporting.
- Familiarity with LabVIEW from academic training; ready to reactivate and deepen it in an industrial test-bench context.
- Experience with hardware/software interfaces through operational systems connected to field equipment.
- Hands-on exposure to electrical test environments: circuits, communication buses, EV charging stations, soldering, maintenance and BO / BE Essais authorization.
- Ability to define, analyse and document test logic, measurement outputs, anomalies and validation evidence.
- Comfortable communicating technical results to multi-disciplinary teams involving software, hardware, R&D, production and operations.
- Strong fit for industrial and critical-system contexts requiring rigor, autonomy, traceability, reliability and clear reporting.

TEAM MANAGEMENT

- Supervision of 3 Master's thesis interns (M2) during software, modelling and data-analysis activities.
- Coordination of technical activities involving research engineers and technicians on a data processing and validation pipeline for the LISA space mission.
- Experience organizing tasks, supporting contributors, reviewing outputs and maintaining alignment across collaborative technical developments.

LANGUAGES

English ● ● ● ● ●

Spanish ● ● ● ● ●

EDUCATION

- **Test data analysis:** Analysed measurement and simulation outputs to identify abnormal behaviours, noise contributions and degradation sources.
- **Signal processing:** Analysed instrumental and environmental effects affecting measurement quality, sensitivity and stability.
- **Noise diagnostics:** Developed methods for spectral analysis, noise characterization, sensitivity studies and cross-sensor correlations.
- **Performance analysis:** Identified contributors to performance degradation and supported prioritisation of correction or mitigation actions.
- **Data-driven decisions:** Analysed measurement and simulation outputs to extract meaningful conclusions from complex technical datasets.
- **Reproducibility:** Produced traceable, reproducible and documented analyses for international scientific and engineering stakeholders.
- **Communication:** Communicated complex results through structured reporting, presentations and collaborative review.

Junior R&D Engineer - Instrumental Data, Simulation & Signal Analysis

ARTEMIS, Observatoire de la Côte d'Azur

📅 2018 – 2021

📍 Nice, France

Context: Data-analysis and simulation activities for gravitational-wave mission preparation, involving weak-signal detection, modelling, simulation and noise characterization.
Objective: Develop robust analysis methods for complex signals and support technical investigations in demanding system environments.

- **Signal analysis:** Developed methods for the separation of overlapping low-SNR signals in complex datasets.
- **Simulation workflows:** Built simulation approaches for complex signal environments, uncertainty studies and large-scale datasets.
- **Algorithm assessment:** Compared simulated outputs, model behaviours and analysis results to assess consistency, robustness and performance limitations.
- **Technical investigations:** Investigated noise sources through cross-sensor correlation analyses in diagnostic contexts.
- **Scientific computing:** Developed strong expertise in Python-based analysis, modelling, documentation and reproducible workflows.

PROFESSIONAL TRAINING

LabVIEW – Academic Training

Engineering / Scientific Instrumentation Training

📅 During academic curriculum

Supervised Machine Learning (Regression & Classification)

DeepLearning.AI - Andrew Ng

📅 2020

📍 Coursera

Recherche reproductible : principes méthodologiques pour une science transparente

FUN MOOC – Inria

📅 2025

📍 France Université Numérique

Continuous Professional Training – Software, Data, AI and Systems

OpenClassrooms

📅 2024 – 2026

Python, SQL, Machine Learning, AI, Linux, JavaScript, TypeScript, Angular, Node.js, Django, REST APIs, C/C++, Java, algorithms and software engineering fundamentals.

PROJECTS

CNES / LISA – Test Logic, Signal Processing & Validation Workflows

ESA/NASA Space Mission - Large-scale International Collaboration

📅 2023 – 2025

📍 Nice, France

Contribution to Python-based signal-processing, model-comparison and validation workflows for the LISA space mission within a large international engineering and scientific collaboration.

Test means relevance: Automated validation logic, repeatable comparison sequences, consistency checks, traceable outputs, technical documentation and review material.

Instrumentation relevance: Signal/noise analysis, sensitivity reasoning, weak-signal interpretation, performance diagnostics and measurement-quality-oriented analysis.

Critical systems relevance: Mission-level performance analysis, traceability, technical evidence, rigorous validation and work in a high-constraint international aerospace context.

Einstein Telescope / LIGO–Virgo–KAGRA – Statistical Diagnostics & Detector Performance

International Collaborations

📅 2021 – 2023

📍 Antwerp, Belgium

Contribution to collaborative developments in data analysis, signal characterization, statistical diagnostics and performance investigations within large-scale international scientific and engineering collaborations.

Role: Development of analysis tools, contribution to technical studies, reporting and collaborative review in high-standard environments.

PhD in Physics

Université Côte d'Azur

📅 2018 – 2021

📍 Nice, France

Selective Graduate Program in Fundamental Physics (Magistère)

Université Paris-Saclay

📅 2016 – 2018

📍 Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

📅 2017 – 2018

📍 Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

📅 2015 – 2016

📍 Orsay, France

System impact: Studies on environmental and instrumental noise sources supporting the identification of performance limitations and design constraints for next-generation gravitational-wave observatories.

VEV – Hardware-Connected Software Services & Electrical Test Support

EV Charging Infrastructure / CPMS Platform

📅 2025 – 2026 📍 Valbonne, France

Contribution to software development, operational diagnostics and hardware/software interface analysis for a CPMS platform connected to EV charging infrastructure.

Industrial testing relevance: Interface diagnostics, data-flow monitoring, incident analysis, software reliability, electrical test support, charging-station maintenance, communication buses, soldering and operational traceability.

STRENGTHS

Test means, instrumentation & acquisition

- Test Means
- Test Benches
- Instrumentation
- LabVIEW
- TestStand Basics
- DAQ
- Instrument Control
- Measurement Chains
- Data Acquisition
- Measurement Analysis
- Automated Test Sequences
- Test Procedure Logic
- Test Sequences
- Automated Testing
- Functional Testing
- Control/Command
- Performance Validation
- Test Logic
- Sensor Data
- Acquisition Quality
- Measurement Reliability
- Calibration Awareness
- Technical Documentation
- Validation Reports

Signal, diagnostics & complex data

- Signal Processing
- Weak Signal Detection
- Low-SNR Analysis
- Noise Modelling
- Noise Diagnostics
- Spectral Analysis
- Cross-Sensor Correlation
- Anomaly Detection
- Pattern Recognition
- Simulation Workflows
- Synthetic Data
- Uncertainty Analysis
- Sensitivity Analysis
- Performance Assessment
- Complex Data Analysis
- Scientific Computing

Hardware/software interfaces & electrical testing

- Hardware/Software Interfaces
- Electrical Test Support
- EV Charging Stations
- Electrical Circuits
- Analog/Digital Electronics Awareness
- Communication Buses
- Bus Diagnostics
- Soldering
- Electrical Checks
- Maintenance
- Integration Tests
- B0 / BE Essais
- Operational Infrastructure
- Field Equipment
- Industrial Diagnostics
- Production Support

Programming, frameworks & tools

- Python
- LabVIEW
- Matlab
- C
- C++
- Java
- Pandas
- NumPy
- Scikit-Learn
- SQL
- Bash
- Linux
- Git
- GitLab
- GitHub
- Docker
- CI/CD
- Jupyter
- REST API
- TypeScript
- Angular
- Node.js

Validation, quality & critical systems

- System Validation
- IVVQ
- V-Cycle
- Verification
- Validation Evidence
- Traceability
- Requirements Awareness
- Technical Evidence
- Reliability
- Robustness
- Risk Identification
- Root-Cause Analysis
- Data Reliability
- Critical Systems
- Aerospace Context
- Space Systems

Collaboration, delivery & communication

- Technical Workshops
- Client-Facing Communication
- Scientific Communication
- Technical Reporting
- Knowledge Sharing
- Stakeholder Alignment
- Cross-Functional Coordination
- Agile Environment
- Confluence
- JIRA
- Zenhub
- OpenSearch
- Sentry
- Technical English
- Autonomy
- Initiative
- Problem Solving
- Analytical Thinking